Got it — here is a **single, unified product + engineering scope** combining:

- **Stage 1** → Visa intelligence
- **Stage 2** → Full execution system

This is written so a **developer or small team can build it end-to-end**.

🧠 PRODUCT OVERVIEW

Product:

AI-powered U.S. visa platform (analysis → full application builder)

User journey:

1. User signs up
2. Completes questionnaire
3. Gets preview
4. Pays $50 → gets full visa strategy
5. Upgrades to $150 → builds full application package
6. Downloads + prints + sends to USCIS (United States Citizenship and Immigration Services)

🧱 SYSTEM ARCHITECTURE (COMBINED)

```
Frontend (Next.js)
        ↓
Backend API (FastAPI)
        ↓
Core Services:
  - Auth Service
  - Profile Service
  - Eligibility Engine
  - Scoring Engine
  - RAG Service
  - LLM Service
  - Case Builder Engine (Stage 2)
  - Document Engine (Stage 2)
  - Form Engine (Stage 2)
        ↓
Databases:
  - PostgreSQL
  - Vector DB (Pinecone)
```

```
        ↓
Storage:
  - S3 (documents, PDFs)
```

⚙️ TECH STACK

Frontend

- Next.js
- Tailwind
- Axios / React Query

Backend

- Python (FastAPI)
- PostgreSQL
- Redis (optional)

AI

- OpenAI (LLM + embeddings)
- Pinecone (RAG)

Payments

- Stripe

🧾 DATABASE SCHEMA (FINAL STRUCTURE)

Core

users

- id
- email
- password_hash
- created_at

user_profiles (versioned)

- id
- user_id

- version
- data (JSONB)
- created_at

Visa System

visas

- id
- code (O1, E2, L1)
- name
- description

visa_rules

- id
- visa_id
- type (eligibility/scoring)
- config (JSONB)
- version

Analysis Layer (Stage 1)

eligibility_results

- id
- user_profile_id
- visa_id
- is_eligible
- reasons

visa_scores

- id
- user_profile_id
- visa_id
- score
- breakdown
- model_version

ai_outputs

- id
- user_profile_id
- type (analysis/strategy)

- prompt
- response
- model

Payments

payments

- id
- user_id
- amount
- status
- stripe_session_id

Case System (Stage 2)

cases

- id
- user_id
- visa_id
- stage (stage1 / stage2)
- progress
- status

case_steps

- id
- case_id
- step_name
- status
- required
- metadata

documents

- id
- case_id
- type
- content
- file_url
- status

forms

- id
- case_id
- form_type
- data (JSONB)
- file_url

🔌 API ENDPOINTS (FULL SET)

AUTH

- POST `/auth/register`
- POST `/auth/login`
- GET `/auth/me`

PROFILE

- POST `/profile`
- GET `/profile/latest`

STAGE 1 — ANALYSIS

Preview (before payment)

- POST `/analysis/preview`

Returns:

- top visa
- partial score

Full analysis (after $50)

- POST `/analysis/full`

Flow:

1. eligibility
2. scoring
3. RAG
4. LLM

PAYMENTS

- POST `/payment/create-session`
- POST `/payment/webhook`

STAGE 2 — CASE SYSTEM

Upgrade user

- POST `/case/upgrade`

Create case

- POST `/case/create`

Get case

- GET `/case`

Checklist (dynamic)

- GET `/case/steps`
- PATCH `/case/steps/:id`

Validate case

- POST `/case/validate`

DOCUMENTS

Generate one

- POST `/documents/generate`

Generate all

- POST `/documents/generate-all`

Get document

- GET `/documents/:id`

FORMS

Autofill form

- POST `/forms/autofill`

Get form

- GET `/forms/:id`

FINAL PACKAGE

Generate package

- POST `/case/generate-package`

Filing instructions

- GET `/case/filing-instructions`

🧠 CORE LOGIC FLOWS

STAGE 1 FLOW

```
User submits questionnaire
   ↓
Save profile
```

```
   ↓
/analysis/preview (free)
   ↓
User pays $50
   ↓
/analysis/full
   ↓
Return:
  - visa ranking
  - score
  - risks
  - strategy
```

STAGE 2 FLOW

```
User upgrades
   ↓
Create case
   ↓
Generate checklist
   ↓
User completes steps (later you define)
   ↓
Generate documents
   ↓
Autofill forms
   ↓
Validate case
   ↓
Generate final package
   ↓
Provide USCIS instructions
```

🤖 AI ARCHITECTURE

1. Eligibility (NO AI)

- Rule-based

2. Scoring (NO AI)

- Deterministic formulas

3. RAG

- Pull USCIS + legal docs

4. LLM

Used for:

- Explanation
- Strategy
- Documents

🎨 FRONTEND SCREENS (FULL FLOW)

1. Landing page

CTA → Start assessment

2. Questionnaire

Multi-step form

3. Preview screen

- show top visa
- blur details

4. Paywall ($50)

5. Results dashboard

- visa ranking
- score
- risks

6. Upgrade screen ($500)

7. Case dashboard

- progress
- checklist
- documents
- forms

8. Document center

- view/edit/generate

9. Final package screen

- download
- print

🔐 COMPLIANCE (MUST IMPLEMENT)

- “Not legal advice”
- Log all AI outputs
- Track model version
- No guarantees

🚀 MVP BUILD PRIORITY (IMPORTANT)

Phase 1 (Launch fast)

- Auth
- Questionnaire
- Analysis (basic)
- Paywall

Phase 2

- RAG
- Better scoring
- Dashboard

Phase 3 (Stage 2 lite)

- Case creation
- Document generation
- PDF export

Phase 4 (Full Stage 2)

- Forms
- Validation
- Final package

🔥 FINAL INSIGHT

This is NOT just an AI product.

You are building:

👉 **Immigration operating system**

Stage 1:
→ sells insight

Stage 2:
→ sells execution